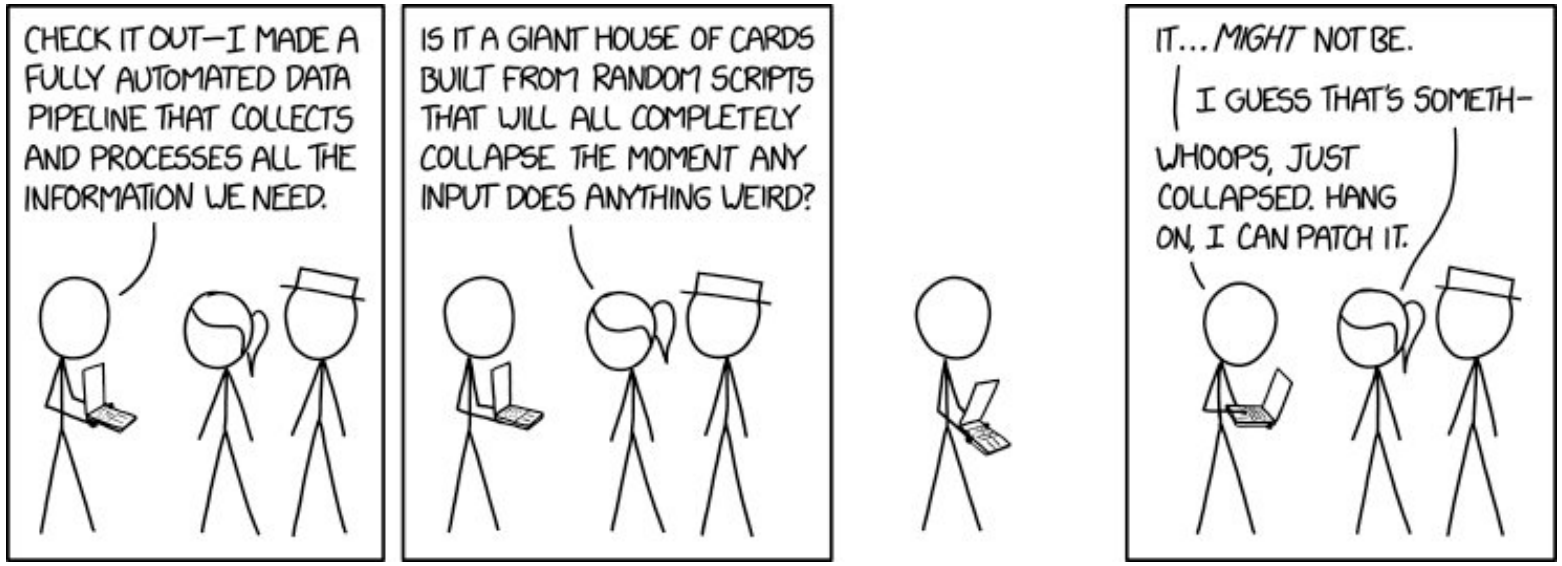
Data Acquisition/Collection

hoarding trash in the internet so cleverly
that we call it an executive strategy

https://github.com/jendralhxr/introdatascience

Data Collection, definition

Process of gathering, measuring, and retrieving raw, unstructured, or structured data from various sources to answer specific research questions, train algorithms, evaluate outcomes, or drive decision-making.



Does the data exist yet?

- Already exists (collection, gathering)
- Exist by other external parties (retrieval)
- Generated through active experiment and survey (measuring)

Method

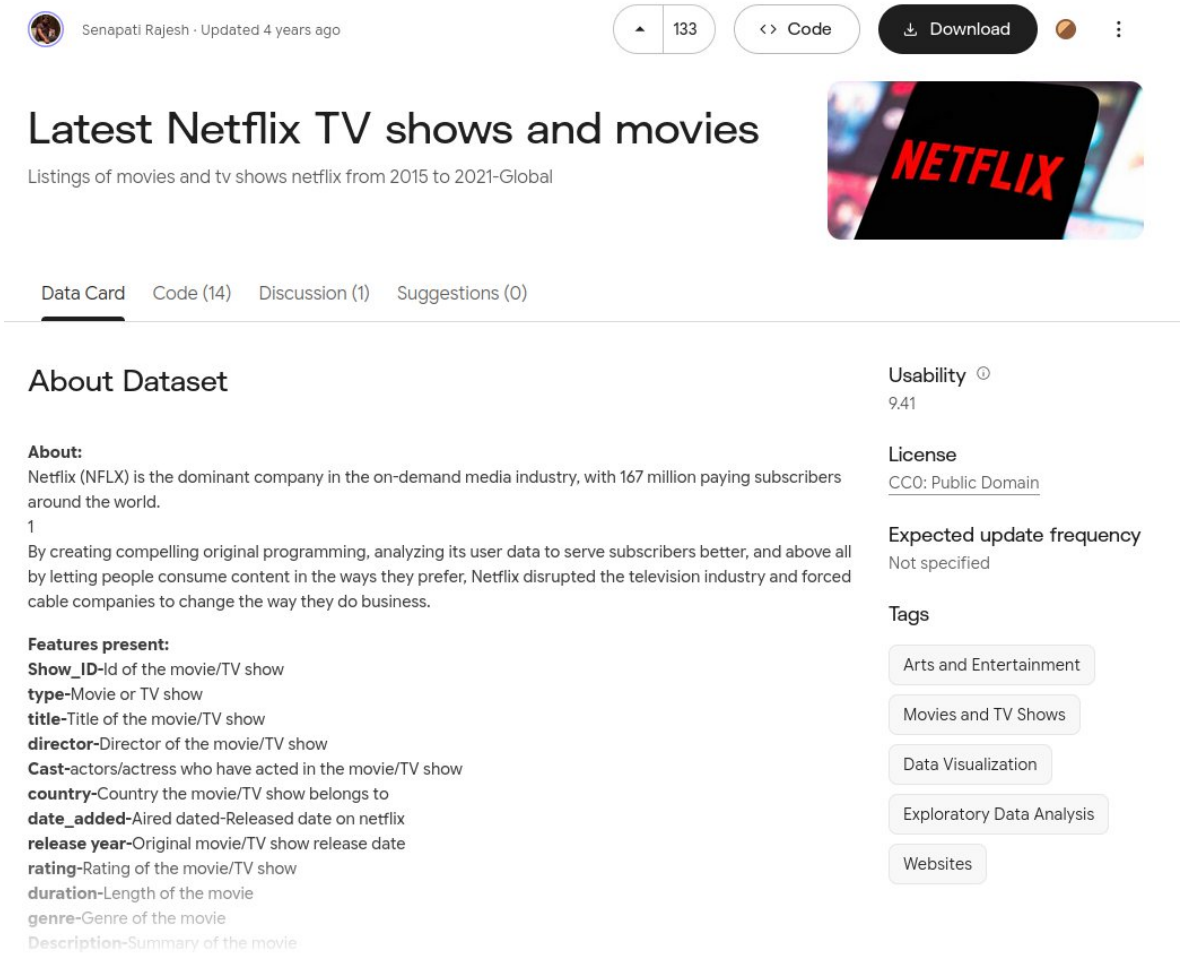
- Dataset initiatives
- Physical hardware sensors
- Controlled experiment research
- Direct human input (questionnaire)
- APIs (Application Programming Interface) and SQL query
- Automated retrieval/scraping
- Synthetic

Attendance List



https://forms.gle/q7yN33EmTYT5dpLB6

Datasets, sb already did the curation



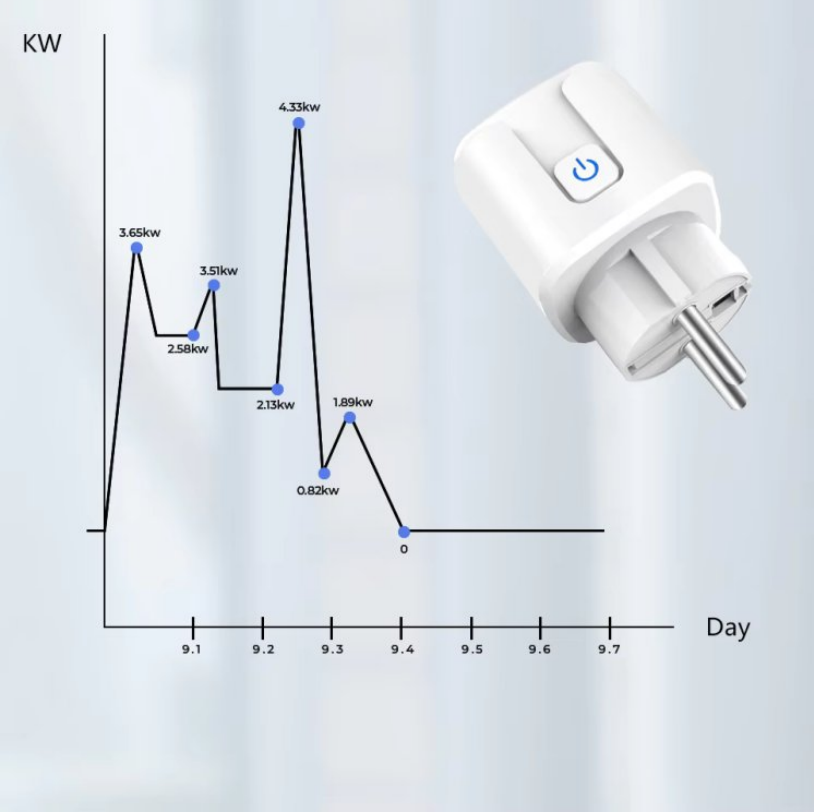

```
import pandas as pd
import kaggle

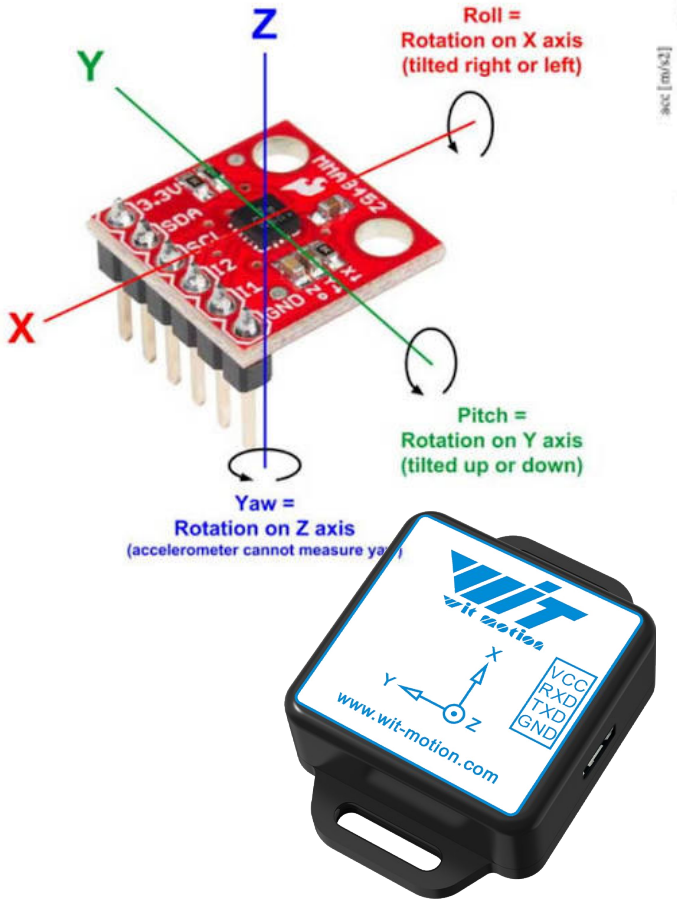
# Authenticate & search public datasets on Kaggle
kaggle.api.dataset_download_files(
    "senapatirajesh/netflix-tv-shows-and-movies", path="./data",
unzip=True)

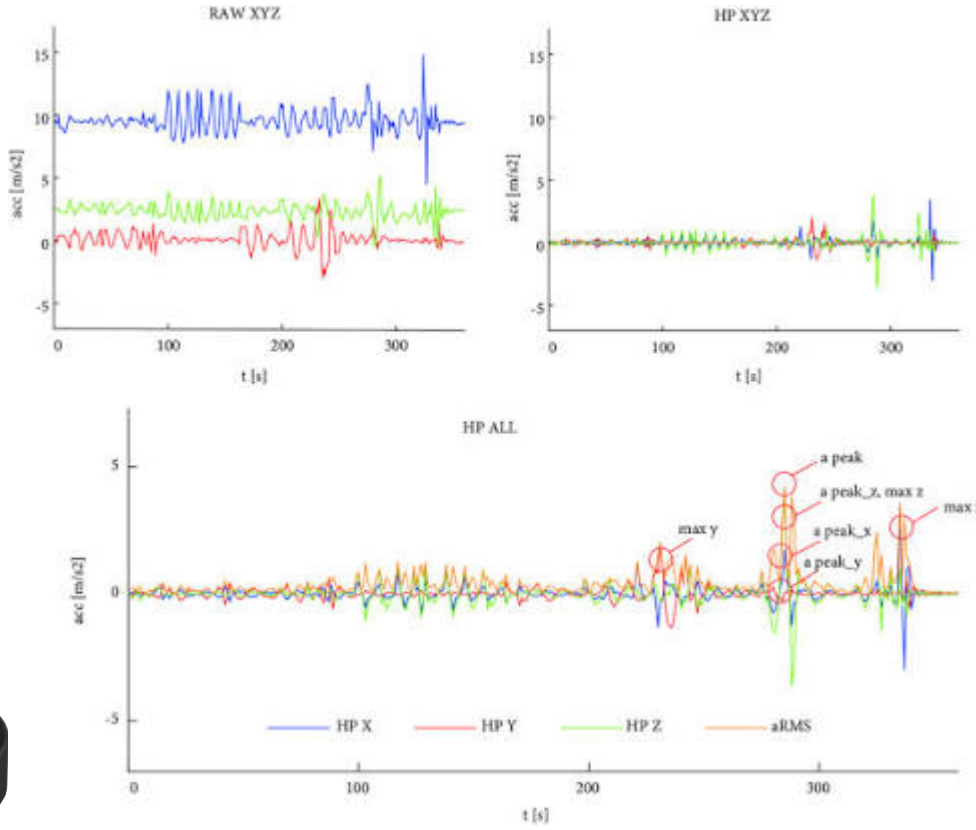
# Load into DataFrame
df = pd.read_csv("./data/NetFlix.csv")
print(df.head())
```

https://www.kaggle.com/datasets/shivamb/netflix-shows

Sensors

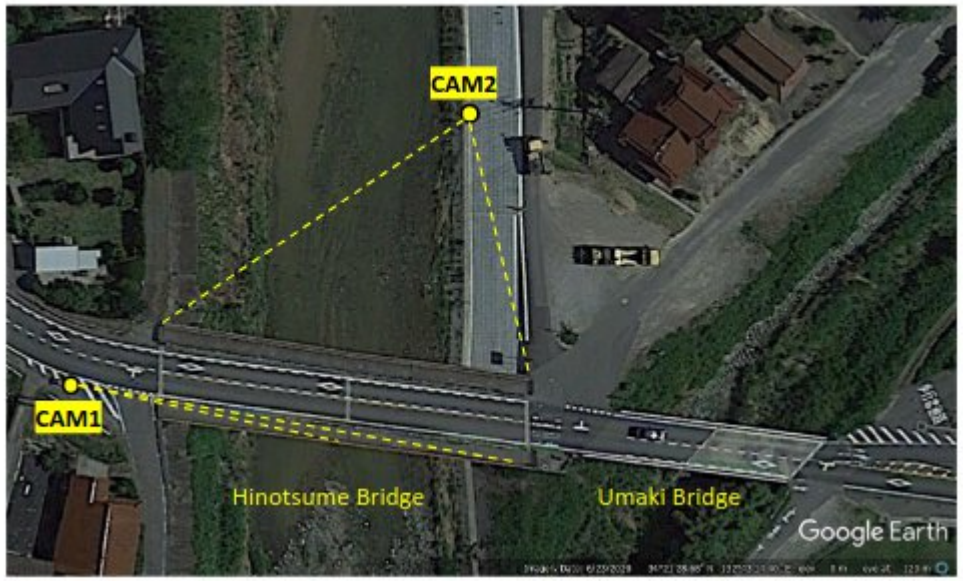




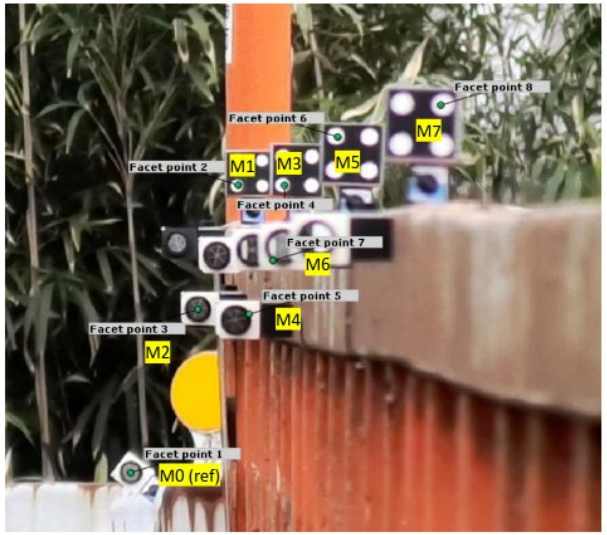


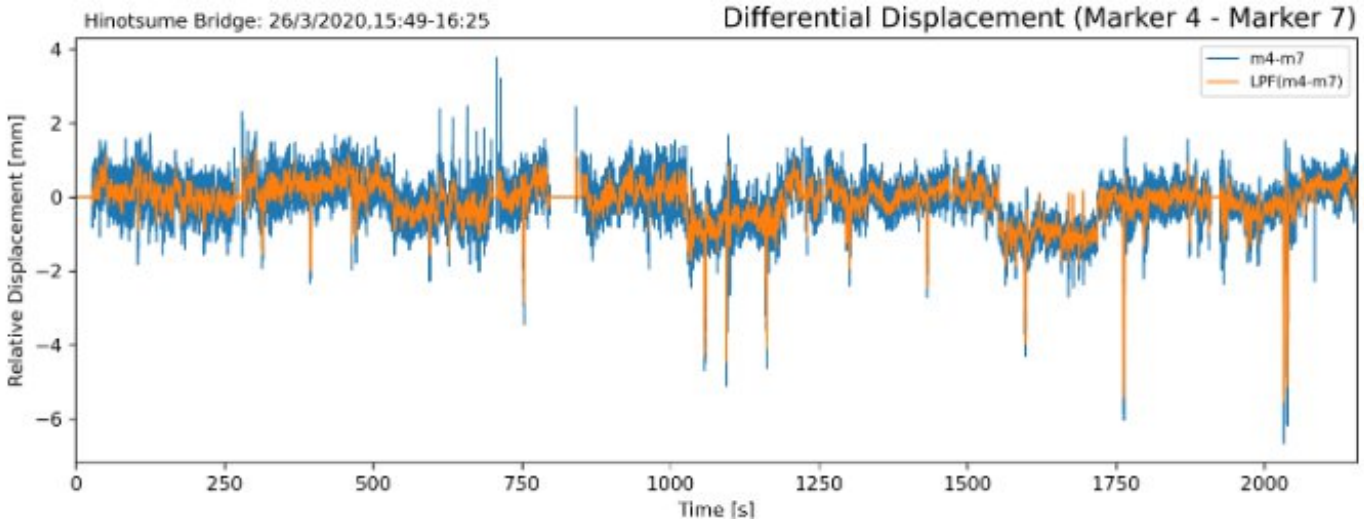
https://www.researchgate.net/publication/332029497_Patient_comfort_level_prediction_during_transport_using_artificial_neural_network/

Research/Experiments







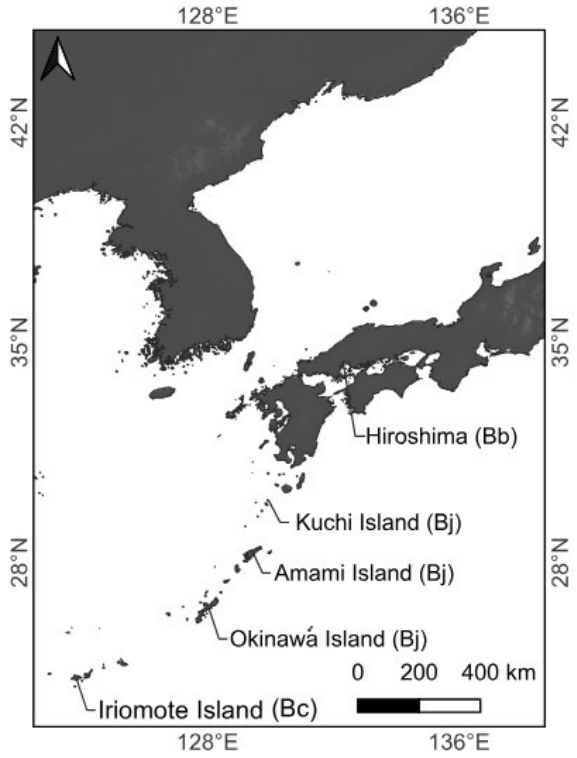


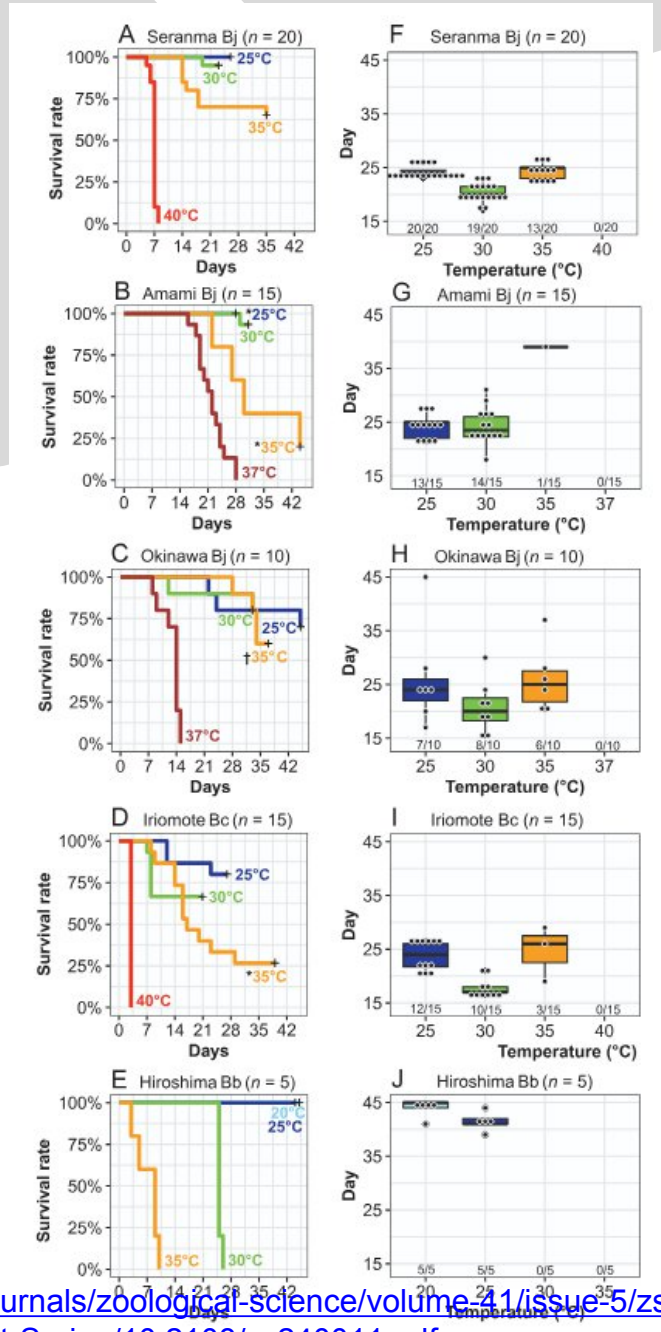
Measuring Bridge vibration when cars/trucks pass

https://www.mdpi.com/1424-8220/21/13/4517

Cont'd







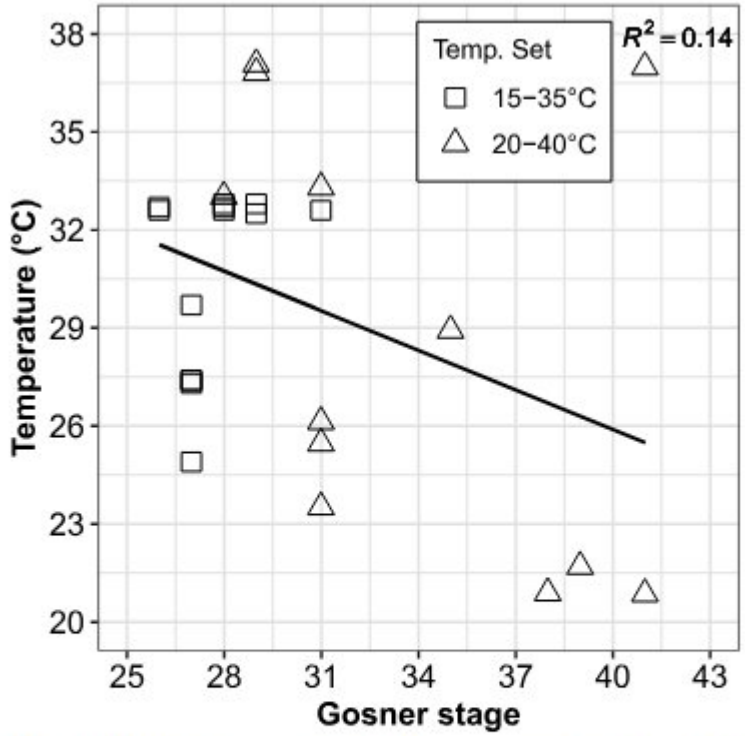


Fig. 5. Preferred temperatures of *B. japonica* from Seranma hot spring (Kuchi Island). Two temperature sets were used: open squares represent 15–35°C, open triangles represent 20–40°C. The regression line was generated using the "lm" method from the *geom_smooth* function within the "*ggplot*" library in RStudio. Details of the temperature preferences data are available in Supplementary Table S3.

Frog tolerance to heat

https://bioone.org/journals/zoological-science/volume-41/issue-5/zs240011/Long-Term-Heat-Tolerance-and-Accelerated-Metamorphosis--Hot-Spring/10.2108/zs240011.pdf

Surveys, questionnaire

Instructions

1. Which of the following activities occupied you on your visit to the unisza Library today? Please check all that apply. Then, check the appropriate box beside each activity to show how successful you were in achieving your aim.

Activity	*Very unsuccessful*	*Fairly unsuccessful*	*Neither successful nor unsuccessful*	*Fairly successful*	*Very successful*
❑ Looked for library materials on the Shelves……	❑	❑	❑	❑	❑
❑ Used an electronic information service (e.g. OPAC, the Internet)….	❑	❑	❑	❑	❑
❑ Used the library's catalogue……..	❑	❑	❑	❑	❑
❑ Got or sought help from library staff	❑	❑	❑	❑	❑
❑ Studied my lessons……	❑	❑	❑	❑	❑
❑ Others, please specify: _____	❑	❑	❑	❑	❑
❑ Others, please specify: _____	❑	❑	❑	❑	❑

2. Approximately, how often do you visit the unisza Library?

❑ Daily ❑ Weekly ❑ Bi-monthly ❑ Monthly

3. What type of materials do you mainly visit the unisza Library to obtain?

❑ Books ❑ Theses
❑ Newspaper/Clippings ❑ Electronic References/Information

https://www.scribd.com/document/49247444/25760632-Library-Services-Satisfaction-Survey-Questionnaire

APIs

data.gov.my Home Dashboards Data Catalogue API Docs Community FAQ Legacy Site

On this page

- Table & Charts
- Metadata
  - Methodology
  - Caveats
  - Publications
  - Variables
  - Next update
  - License
- Download
- Programmatic Access
  - Full dataset
  - Open API

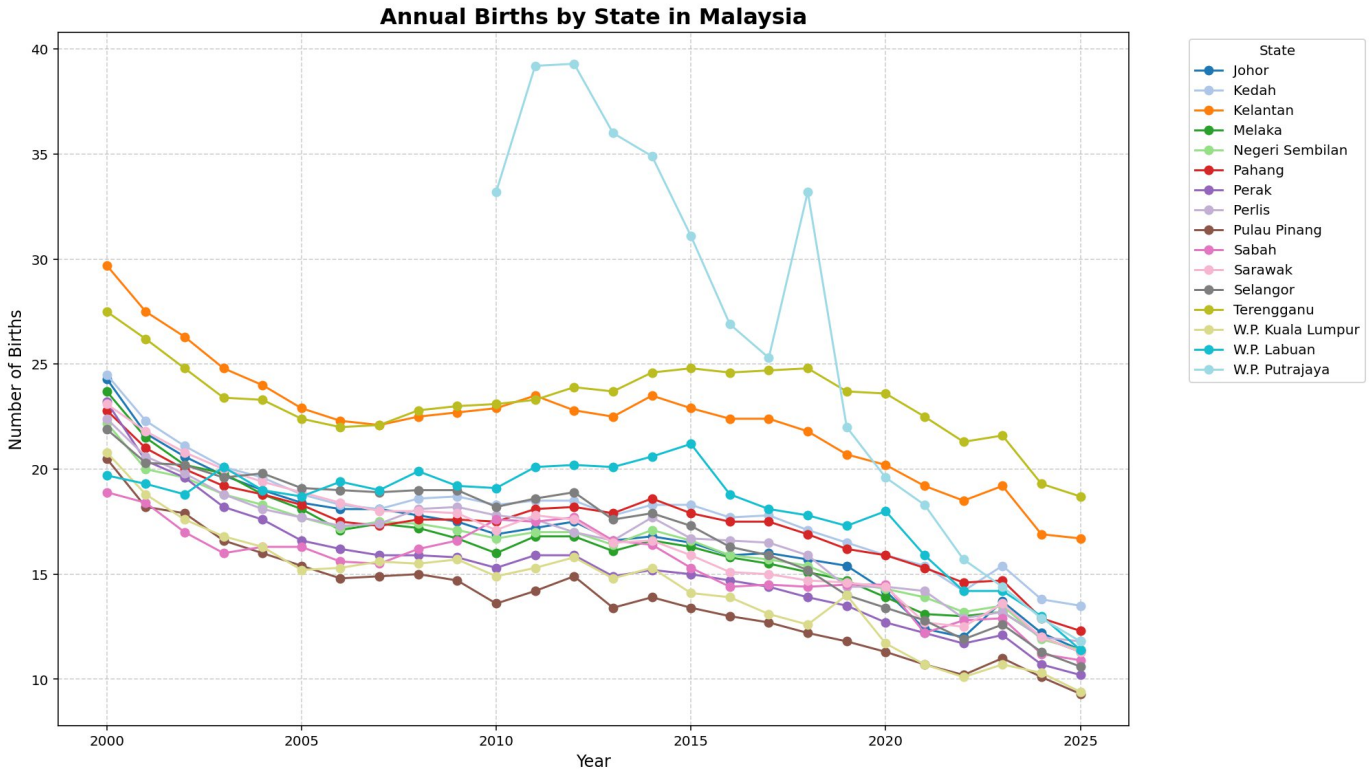
Annual Live Births by State

Annual number of births with signs of life upon delivery, by state.

Johor

Search

State	Year	Live Births	Crude Birth Rate
Johor	2000-01-01	67,183	24.3
Johor	2001-01-01	61,432	21.7
Johor	2002-01-01	59,699	20.6
Johor	2003-01-01	58,230	19.7
Johor	2004-01-01	57,339	19
Johor	2005-01-01	56,524	18.4
Johor	2006-01-01	56,737	18.1
Johor	2007-01-01	57,780	18.1
Johor	2008-01-01	57,838	17.8
Johor	2009-01-01	57,849	17.5
Johor	2010-01-01	56,923	16.9
Johor	2011-01-01	58,868	17.2
Johor	2012-01-01	61,187	17.5
Johor	2013-01-01	58,855	16.6
Johor	2014-01-01	61,307	16.8



APIs, cont’d

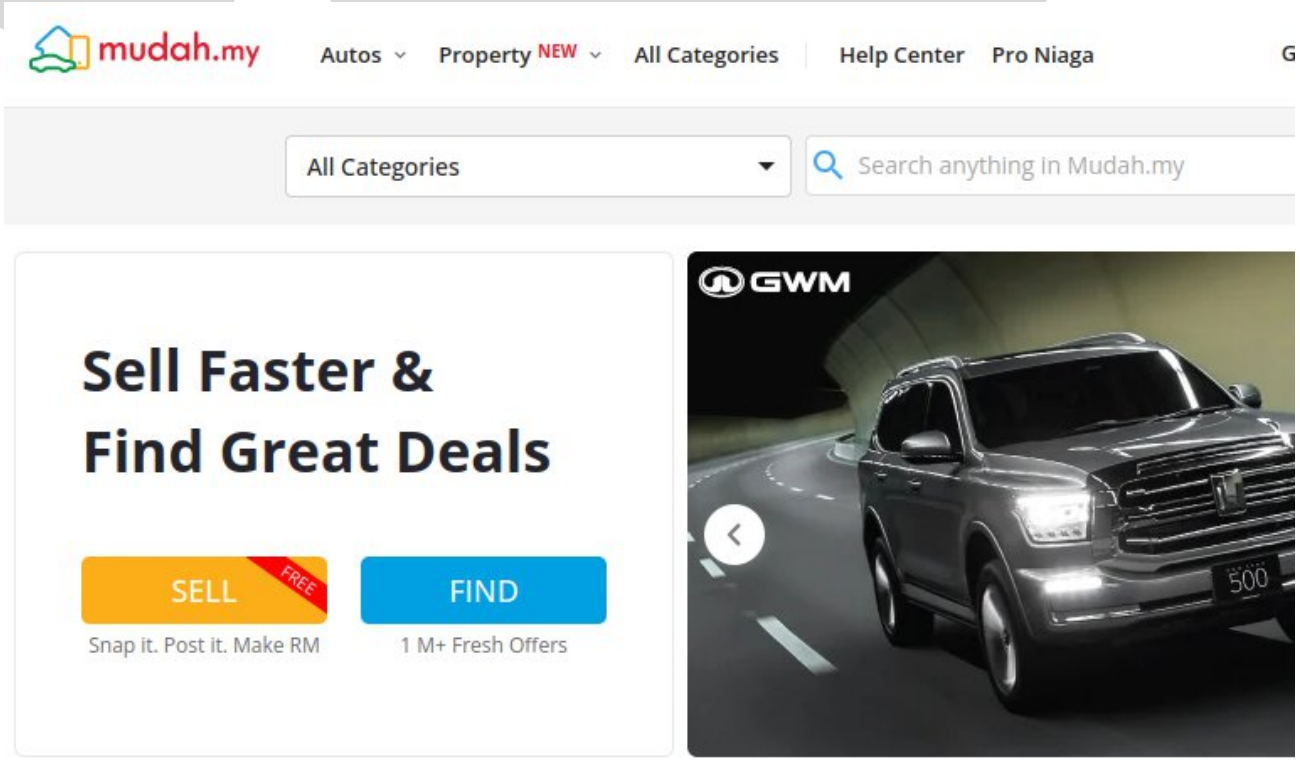
```
import matplotlib.pyplot as plt
import pandas as pd
import requests

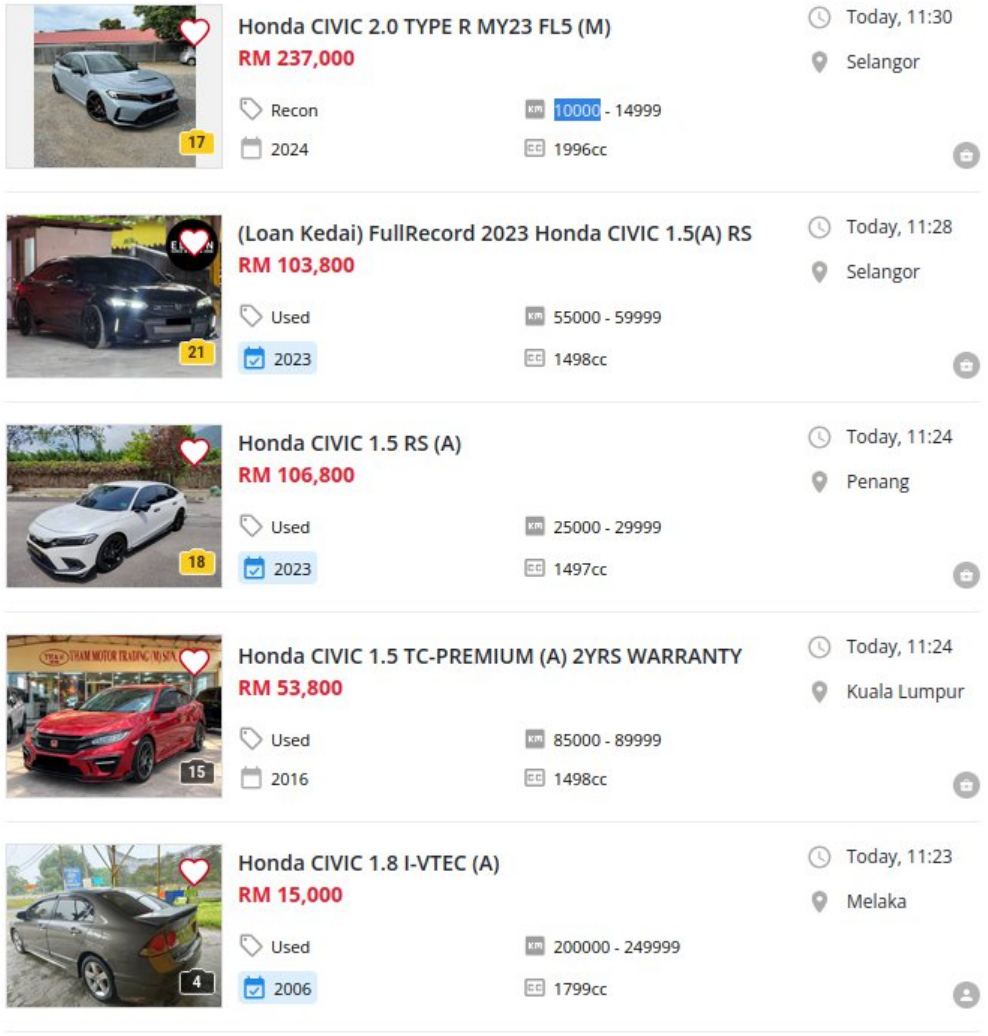
# Fetch data directly from data.gov.my API
API_URL = "https://api.data.gov.my/data-catalogue?id=births_annual_state"
response = requests.get(API_URL)
data = response.json()
df = pd.DataFrame(data)
df["date"] = pd.to_datetime(df["date"])
df["year"] = df["date"].dt.year
df["births"] = pd.to_numeric(df["rate"])
df_pivot = df.pivot(index="year", columns="state", values="births")

# Plotting with a Distinct Color for Each State
plt.figure(figsize=(14, 8))
ax = df_pivot.plot(kind="line", marker="o", figsize=(14, 8), cmap="tab20")
plt.title("Annual Births by State in Malaysia", fontsize=16, fontweight="bold")
plt.xlabel("Year", fontsize=12)
plt.ylabel("Number of Births", fontsize=12)
plt.grid(True, linestyle="--", alpha=0.6)
plt.legend(
    title="State", bbox_to_anchor=(1.05, 1), loc="upper left", frameon=True
)
ax.yaxis.set_major_formatter("{x:,.0f}")

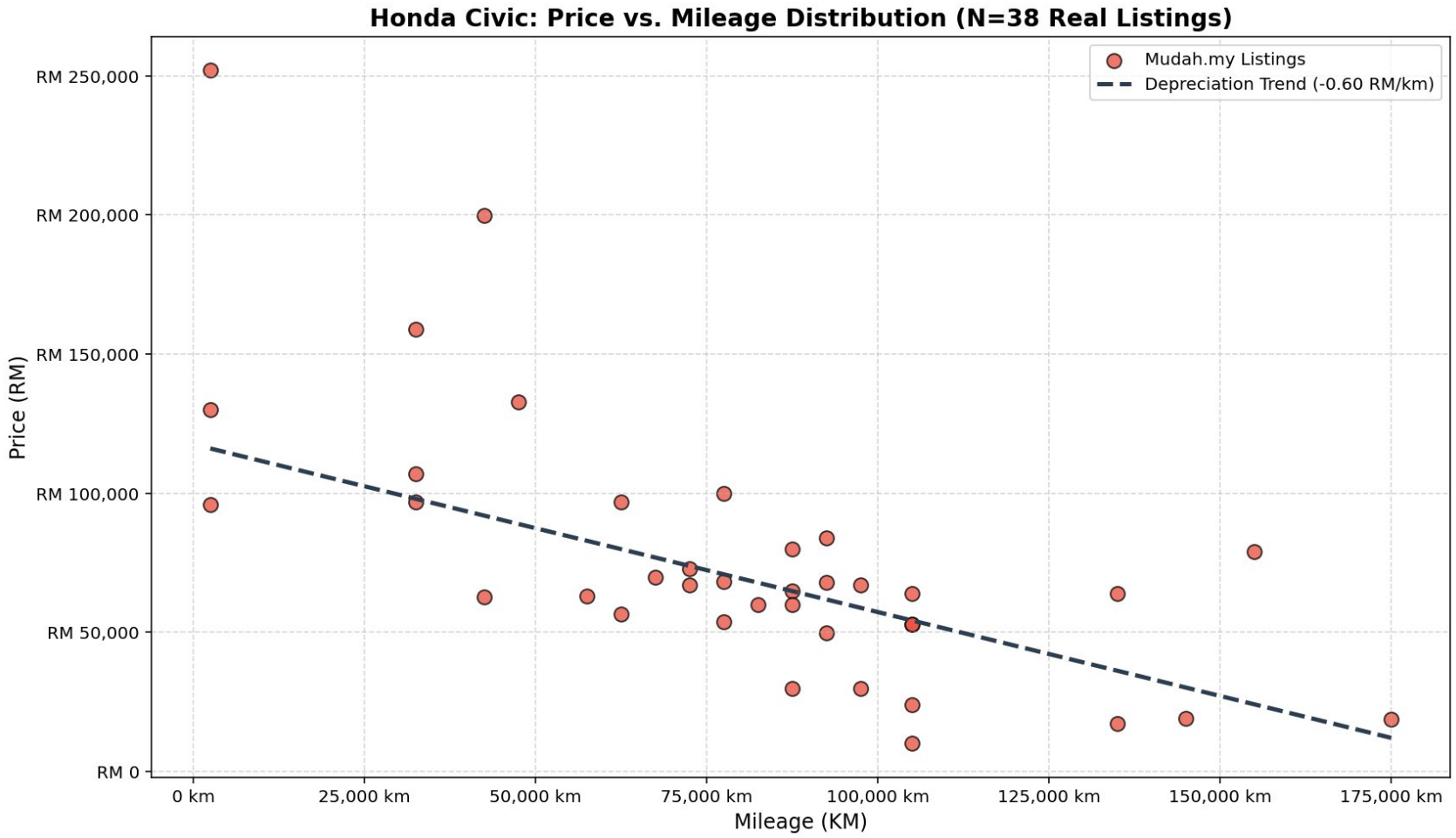
plt.tight_layout()
plt.show()
```

Web scraping



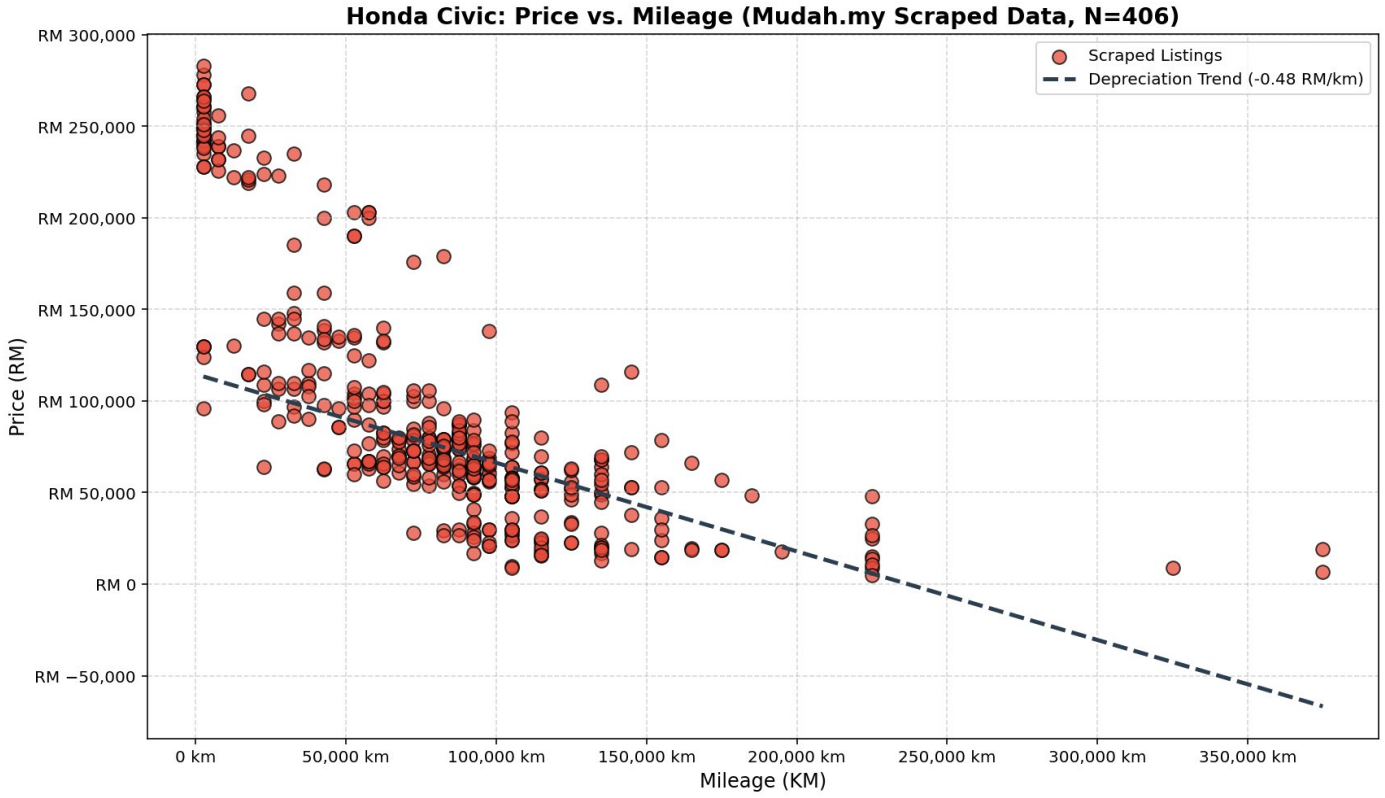


```
...
16
Honda CIVIC 1.5 TC-P (A)
RM 79,800
Mudah Certified
Used
85000 - 89999
2020
1498cc
Today, 01:09
Selangor
12
Honda CIVIC 1.5 TC (A)
RM 66,888
Mudah Certified
Used
95000 - 99999
2018
1498cc
Today, 00:30
Selangor
22
Honda CIVIC 1.5 TC (A)
RM 78,800
Mudah Certified
Used
150000 - 159999
2021
1498cc
Today, 00:30
Kuala Lumpur
2
hotwheels honda civic
...
```



Automated, Scaled up

```
Fetching Page 1...
curl_cffi fetch failed on page 1: Session.request()
got an unexpected keyword argument 'follow_redirects'
  -> Captured 38 valid Civic listings.
Fetching Page 2...
curl_cffi fetch failed on page 2: Session.request()
got an unexpected keyword argument 'follow_redirects'
  -> Captured 38 valid Civic listings.
...
...
Fetching Page 19...
curl_cffi fetch failed on page 19: Session.request()
got an unexpected keyword argument 'follow_redirects'
  -> Captured 40 valid Civic listings.
Fetching Page 20...
curl_cffi fetch failed on page 20: Session.request()
got an unexpected keyword argument 'follow_redirects'
  -> Captured 36 valid Civic listings.
```



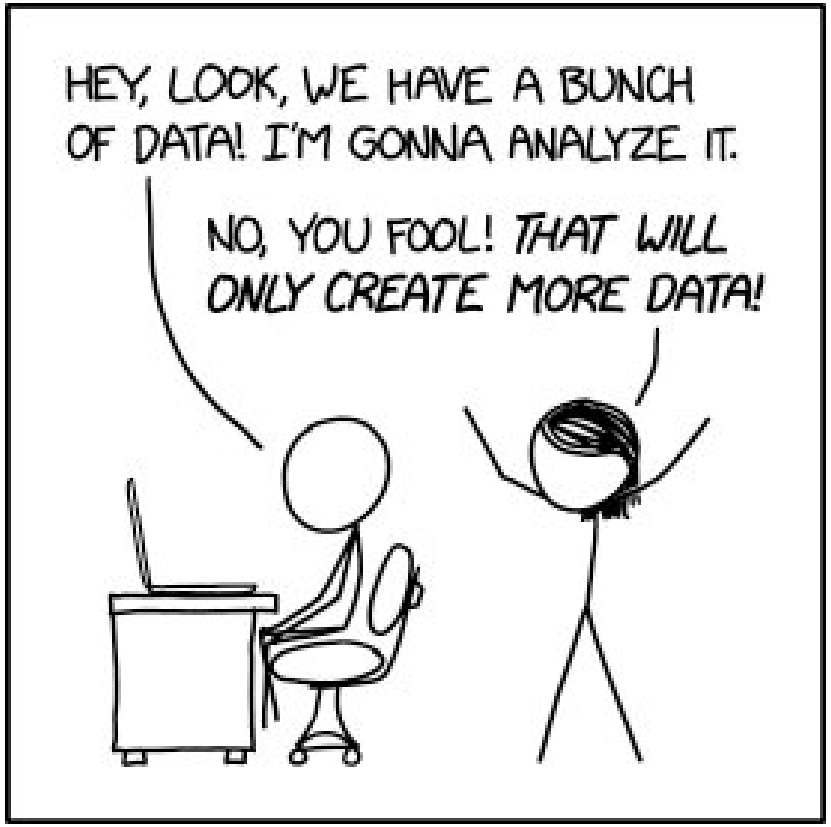
Synthetic

give me 6 images of cat



Wrap up

- Data is already everywhere
- But if one needs specific data, they need to get their hands dirty
- Parsing the information and understanding the context is the utmost important



https://xkcd.com/2582/

Assignment



- Fetch data from data.gov.my to your liking
- Plot some graph based on the data
- Give some explanation/discussion, preferably with supporting evidences
- Submit as PDF file (including the python code, the graph, and the discussion)
- Send over to muhammad_zulhaj.sada@upnjatim.ac.id by **2026-10-30 1 PM** Malaysia time, late submission will have score deducted